Source: **Fraunhofer IIS**

Title: **ATIAS Contents**

Document for: **Discussion and Agreement**

Agenda Item: **Unique**

Summary

This contribution aims at triggering a discussion on the contents for the two specifications 26.261 (requirements) and 26.260 (objective test methods) targeted by ATIAS. The general presentation scheme is to provide sections (with content, where applicable, or just the heading) following the structure of 26.131/132 merged to a single document, regardless of later on distribution over several specifications.

Section W contains a proposal on “Interfaces” that could go to section 4 of the draft TS 26.261. The text has been derived from TS 26.132 and adapted for access technologies of relevance for immersive audio services plus an editor’s note to discuss the potential acoustic interfaces.

Section X contains a proposal on “Performance for conversational services” that could go to section 5 of the draft TS 26.261. It is proposed to discuss the test cases copied from TS 26.131 for their applicability to immersive conversational services. It should be noted that the list may not be exhaustive for immersive conversational services with new challenges imposed by spatial audio capturing and playback.

Section Y contains a proposal on “Test Configurations” that could go to a draft CR to 26.260. Some common sections in 26.132 seem to be still applicable, with minor edits required. Editor’s notes are inserted at places where a discussion is required how to derive the test configurations while IVAS is not yet finalized.

Section Z contains a proposal on “Test Methods” that could go to a draft CR to 26.260. The list should essentially mirror the performance requirements section in terms of structure, thus the same discussion applies.

In summary, the source proposes to discuss the contents of this contribution as potential contents for the ATIAS-affected specifications. If there can be consensus derived on certain aspects, it is furthermore proposed to move agreeable contents to the draft TS 26.261 or a new draftCR for TS26.260. A potential alternative is to start a permanent document, which would be later on used for input to the TS and CR26.260.

This contribution is a revision of S4-AHQ144, which has been presented at an SQ conference call. Given that there was good feedback and a good discussion but no concrete proposals for change, the document is resubmitted.

W Interfaces

In TS 26.131 and TS 26.132 the acoustic interfaces, air interface and the point of interconnect (POI) are defined (see Figure 1. From TS 26.132).

NOTE : Connection to PSTN should include electrical echo control (EEC).

Figure 1: Interfaces (MRP, ERP/DRP…, Air interface and Point of interconnect) for specification of terminal acoustic characteristics

Contrary to 26.131/132 the air interfaces for ATIAS should focus on the access technologies LTE, 5G NR, and WLAN access to the EPC, whereas GSM and 3G may be of less relevance for immersive audio services.

The acoustic interfaces in 26.131/132 have been defined as the following (always assuming that a UE may also be a softphone UE):

- Handset UE used as a handset;
- Headset UE used with headset;
- Vehicle Mounted Hands-free UE mounted in a vehicle;
- Desktop-mounted hands-free UE with external loudspeaker(s) used in hands-free mode;
- Hand-held hands-free UE with internal loudspeaker(s) used in hands-free mode.

Editor's Note: Those acoustic interfaces seem to be still of relevance, where for ATIAS there should be a focus on acoustic interfaces that actually allow the capturing or playback of immersive audio. While there may be little improvement possible for a handset UE used as a handset (as this is commonly understood as a mono signal being recorded and being played back monaurally), the other acoustic interfaces already cover capturing or rendering setups with multiple microphones or multiple speakers (though not used for immersive audio). It is also envisioned that additional acoustic interfaces may be defined, as new device types and form factors appear.

X Performance for conversational services

X.1 Applicability

X.2 Overall loss/loudness ratings

X.3 Idle Channel Noise

X.4 Sensitivity/Frequency Characteristics

X.5 Sidetone Characteristics

X.6 Stability Loss

X.7 Acoustic Echo Control

X.8 Distortion

X.9 Delay

X.10 Echo Control Characteristics

X.11 Quality in the presence of ambient noise

X.12 Jitter buffer management behaviour

Y Test Configurations

Y.1 Setup for Terminals

Y.1.1 Common Setup for All Terminals

The preferred acoustic access to LTE, 5G NR, and WLAN terminals is the most realistic simulation of the "average" subscriber. This can be made by using HATS (head and torso simulator), with appropriate ear simulation and appropriate mountings of handset terminals to the HATS in a realistic but reproducible way. Hands-free terminals shall use the HATS or free field microphone techniques in a realistic but reproducible way.

HATS is described in ITU-T Recommendation P.58 [15], appropriate ears are described in ITU-T Recommendation P.57 [14] (Type 3.3), proper positioning of handsets in realistic conditions is found in ITU-T Recommendation P.64, and the test setups for various types of hands-free terminals can be found in ITU-T Recommendation P.581.

Unless stated otherwise, if a volume control is provided, the setting is chosen such that the nominal RLR is met as close as possible.

The preferred way of testing is the connection of a terminal to the system simulator with exact defined settings and access points. The test sequences are fed in either electrically using a reference codec, using the direct signal processing approach, or acoustically using ITU-T specified devices.

The system simulator shall simulate the access network and core network including the immersive audio encoding/decoding specified for the test (i.e. IVAS) but excluding further transcoding beyond linear PCM.

Editor's Note: IVAS may not yet be available for the specification of test configurations. During this period linear PCM coding of the signals should be assumed to derive appropriate requirements.

Unless specified otherwise for the respective test, the radio conditions on the air interface shall have a block error rate of 0% and the jitter in the IP transport for MTSI-based speech shall be ? 1 ms.

NOTE 1: For WLAN connections, an RF shielded room may be one way to achieve these requirements on block error rate and jitter. Otherwise, care should be taken with potential sources of radio interference and their impact.

In case of MTSI-based speech, the reference client shall allow to synchronize to the clock of the device under test and include a de-jitter buffer to equalize possible jitter in the signal received from the UE.

When operating with synchronized clock, the de-jitter buffer shall be a static de-jitter buffer and the jitter buffer management shall not compensate for clock skew. The reference client shall not lose or discard packets, shall not trigger retransmission, and shall not use error concealment or time-warping. The initial jitter buffer size (filling level) shall be higher than the maximum expected network jitter and the maximum jitter buffer size shall be at least twice the initial size. During jitter buffer reset, the de-jitter buffer shall be emptied/filled to the initial buffer size. In case of buffer over- or underruns, the reference client shall give a warning and it shall be reported.

NOTE 2: A static de-jitter buffer is a first-in-first-out (FIFO) buffer which at the beginning buffers packets until a given initial buffer size is reached. Due to changing network delays the filling level of the de-jitter buffer can change, but the sum of network delay and jitter buffer delay is constant (as opposed to an adaptive jitter buffer management). The filling level of the de-jitter buffer represents the de-jitter buffer delay.

For measurements with unsynchronized clock e.g. the measurement of clock skew, jitter buffer over- and underruns cannot be avoided due to the unsynchronized clocks. Under the assumption of jitter-free condition the initial jitter buffer size (filling level) shall be chosen such that the maximum clock skew can be compensated without any loss of packets for a given time. For the measurement of clock skews the jitter buffer size should be chosen such that for clock skew of up to 100ppm no loss of packets due to buffer over- or under-run shall occur for a sequence of 160s.

For LTE connections, the system simulator shall be configured for FDD operation, with a default or dedicated bearer and reference measurement channel scheduling that provides enough resource block allocation for transmitting a full speech packet within a transmission time interval of 1ms. No HARQ re-transmissions shall occur. TDD operation, TTI bundling, connected DRX and other forms of scheduling (e.g. SPS) are for further study.

The test setup has to ensure proper clock synchronization of the test equipment to the UE. Clock skew shall be negligible and packet loss shall not occur during the test.

Y.1.2 Setup for handset terminals

Y.1.3 Setup for headset terminals

Y.1.4 Setup for hands-free terminals

Y.1.4.1 Vehicle-mounted hands-free

Y.1.4.2 Desktop hands-free

Y.1.4.3 Hand-held hands-free

Y.1.4.4 Softphone including speakers and microphone

Y.1.4.4 Softphone with separate speakers

Y.2 Setup of the electrical interfaces

Y.2.1 Codec approach and specification

In this approach, a codec is used to convert the digital input/output bit-stream of the system simulator to the equivalent analogue values. With this approach a system simulator simulating the radio link to the terminal under controlled and error-free conditions is required, unless otherwise specified for the respective test. The system simulator has to be equipped with a high-quality codec with characteristics as close as possible to ideal.

Definition of 0 dBr point:

D/A converter - a Digital Test Sequence (DTS) representing the codec equivalent of an analogue sinusoidal signal with an RMS value of 3,14 dB below the maximum full-load capacity of the codec shall generate 0 dBm across a 600 ohm load;

A/D converter - a 0 dBm signal generated from a 600 ohm source shall give the digital test sequence (DTS) representing the codec equivalent of an analogue sinusoidal signal with an RMS value of 3,14 dB below the maximum full-load capacity of the codec.

Channel-based telephony testing

For testing of a LTE, 5G NR or WLAN terminal supporting multi-channel telephony, the system simulator shall use the IVAS speech codec [configuration TBD …].

Editor's Note: IVAS may not yet be available for the specification of test configurations. During this period linear PCM coding of the signals should be assumed to derive appropriate requirements that may need final adjustment with the final IVAS codec.

Object-based telephony testing

For testing of a LTE, 5G NR or WLAN terminal supporting object-based telephony, the system simulator shall use the IVAS speech codec [configuration TBD …].

Editor's Note: IVAS may not yet be available for the specification of test configurations. During this period linear PCM coding of the signals should be assumed to derive appropriate requirements that may need final adjustment with the final IVAS codec.

Scene-based telephony testing

For testing of a LTE, 5G NR or WLAN terminal supporting scene-based telephony, the system simulator shall use the IVAS speech codec [configuration TBD …].

Editor's Note: IVAS may not yet be available for the specification of test configurations. During this period linear PCM coding of the signals should be assumed to derive appropriate requirements that may need final adjustment with the final IVAS codec.

Y.2.2 Direct digital processing approach

In this approach, the digital input/output bit-stream of the terminal connected through the radio link to the system simulator is operated upon directly.

Channel-based telephony testing

For testing of a LTE, 5G NR or WLAN terminal supporting multi-channel telephony, the system simulator shall use the IVAS speech codec [configuration TBD …].

Editor's Note: IVAS may not yet be available for the specification of test configurations. During this period linear PCM coding of the signals should be assumed to derive appropriate requirements that may need final adjustment with the final IVAS codec.

Object-based telephony testing

For testing of a LTE, 5G NR or WLAN terminal supporting object-based telephony, the system simulator shall use the IVAS speech codec [configuration TBD …].

Editor's Note: IVAS may not yet be available for the specification of test configurations. During this period linear PCM coding of the signals should be assumed to derive appropriate requirements that may need final adjustment with the final IVAS codec.

Scene-based telephony testing

For testing of a LTE, 5G NR or WLAN terminal supporting scene-based telephony, the system simulator shall use the IVAS speech codec [configuration TBD …].

Editor's Note: IVAS may not yet be available for the specification of test configurations. During this period linear PCM coding of the signals should be assumed to derive appropriate requirements that may need final adjustment with the final IVAS codec.

Y.2.4 Position and calibration of HATS

The horizontal positioning of the HATS reference plane shall be guaranteed within ± 2(for testing hands-free equipment.

The HATS shall be equipped with a Type 3.3 Artificial Ear. For hands-free measurements the HATS shall be equipped with two artificial ears. The pinnae are specified in Recommendation P.57 [14] for Type 3.3 artificial ears. The pinnae shall be positioned on HATS according to ITU-T Recommendation P.58 [15].

The exact calibration and equalization procedures as well as how to combine the two ear signals for the purpose of measurements can be found in ITU-T Recommendation P.581.

If not stated otherwise, the HATS shall be diffuse-field equalized using values from ITU-T Recommendation P.58 [15]. For 1/3-octave band measurements, the inverse of the nominal diffuse field curve in P.58 table 3 shall be used. For 1/12-octave band measurements, the inverse of the nominal diffuse field curve in P.58 Annex A shall be used. For measurements requiring diffuse-field correction values for frequencies other than those used in the P.58 tables, linear interpolation on a log frequency scale from the P.58 Annex A values shall be used.

For hand-held hands-free UE, the setup corresponding to ’portable hands-free’ in ITU-T Recommendation P.581 should be used.

Y.3 Test signals

Z Test Methods

Z.1 Applicability

Z.2 Overall loss/loudness ratings

Z.3 Idle Channel Noise

Z.4 Sensitivity/Frequency Characteristics

Z.5 Sidetone Characteristics

Z.6 Stability Loss

Z.7 Acoustic Echo Control

Z.8 Distortion

Z.9 Delay

Z.10 Echo Control Characteristics

Z.11 Quality in the presence of ambient noise

Z.12 Jitter buffer management behaviour